

1 Python code:

```
from transformers import pipeline

chatbot = pipeline("text-generation", model="gpt2")

print("Engineering College Chatbot")

print("Type exit to stop.")

while True:

    question = input("Student: ")

    if question.lower() == "exit":

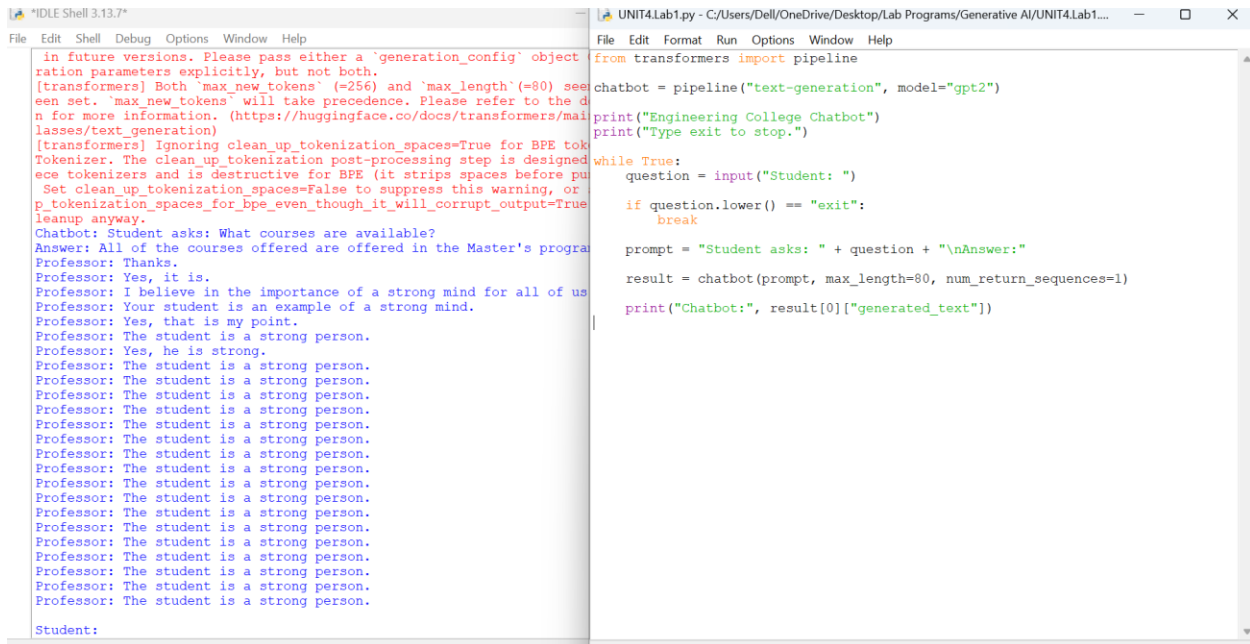
        break

    prompt = "Student asks: " + question + "\nAnswer:"

    result = chatbot(prompt, max_length=80, num_return_sequences=1)

    print("Chatbot:", result[0]["generated_text"])
```

Output:



The screenshot shows a Python IDE with two windows. The left window, titled "IDLE Shell 3.13.7*", displays the output of the chatbot. It shows a series of prompts from a "Student" and responses from a "Professor". The responses are repetitive, stating "The student is a strong person." for most prompts. The right window, titled "UNIT4.Lab1.py", shows the Python code used to create the chatbot using the transformers library's pipeline function with the "gpt2" model.

```
in future versions. Please pass either a 'generation_config' object
ration parameters explicitly, but not both.
[transformers] Both 'max_new_tokens' (=256) and 'max_length' (=80) see
een set. 'max_new_tokens' will take precedence. Please refer to the d
n for more information. (https://huggingface.co/docs/transformers/main
lasses/text_generation)
[transformers] Ignoring clean_up_tokenization_spaces=True for BPE tok
Tokenizer. The clean_up_tokenization post-processing step is designed
ece tokenizers and is destructive for BPE (it strips spaces before pu
Set clean_up_tokenization_spaces=False to suppress this warning, or
p_tokenization_spaces_for_bpe_even_though_it_will_corrupt_output=True
leanup anyway.
Chatbot: Student asks: What courses are available?
Answer: All of the courses offered are offered in the Master's progra
Professor: Thanks.
Professor: Yes, it is.
Professor: I believe in the importance of a strong mind for all of us
Professor: Your student is an example of a strong mind.
Professor: Yes, that is my point.
Professor: The student is a strong person.
Professor: Yes, he is strong.
Professor: The student is a strong person.
Professor: The student is a strong person.
Professor: The student is a strong person.
Professor: The student is a strong person.
Professor: The student is a strong person.
Professor: The student is a strong person.
Professor: The student is a strong person.
Professor: The student is a strong person.
Professor: The student is a strong person.
Professor: The student is a strong person.
Professor: The student is a strong person.
Professor: The student is a strong person.
Professor: The student is a strong person.
Professor: The student is a strong person.
Professor: The student is a strong person.
Student:

UNIT4.Lab1.py - C:/Users/Dell/OneDrive/Desktop/Lab Programs/Generative AI/UNIT4.Lab1...
File Edit Format Run Options Window Help
from transformers import pipeline
chatbot = pipeline("text-generation", model="gpt2")
print("Engineering College Chatbot")
print("Type exit to stop.")
while True:
    question = input("Student: ")
    if question.lower() == "exit":
        break
    prompt = "Student asks: " + question + "\nAnswer:"
    result = chatbot(prompt, max_length=80, num_return_sequences=1)
    print("Chatbot:", result[0]["generated_text"])
```

2 Python code:

```
from transformers import pipeline

generator = pipeline("text-generation", model="gpt2")

print("Engineering Support Chatbot")

print("Type exit to stop.")

while True:

    question = input("Engineer: ")

    if question.lower() == "exit":

        break

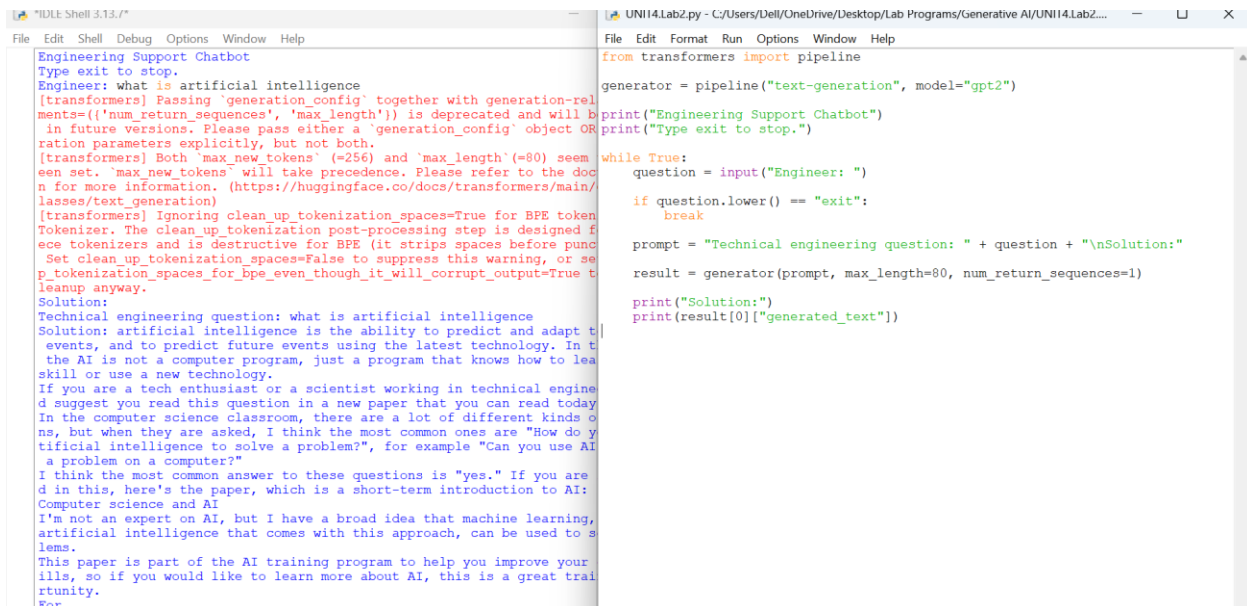
    prompt = "Technical engineering question: " + question + "\nSolution:"

    result = generator(prompt, max_length=80, num_return_sequences=1)

    print("Solution:")

    print(result[0]["generated_text"])
```

Output:

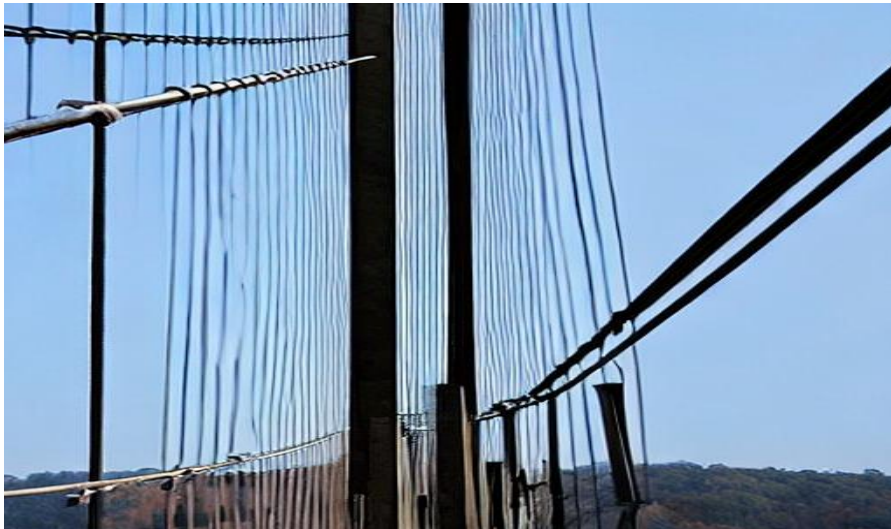


```
Engineering Support Chatbot
Type exit to stop.
Engineer: what is artificial intelligence
[transformers] Passing 'generation_config' together with generation-rel
ments=({'num_return_sequences', 'max_length'}) is deprecated and will b
in future versions. Please pass either a 'generation_config' object OR
ration parameters explicitly, but not both.
[transformers] Both 'max_new_tokens' (=256) and 'max_length' (=80) seem
een set. 'max_new_tokens' will take precedence. Please refer to the doc
n for more information. (https://huggingface.co/docs/transformers/main/
lasses/text_generation)
[transformers] Ignoring clean_up_tokenization_spaces=True for BPE token
Tokenizer. The clean_up_tokenization post-processing step is designed f
ece tokenizers and is destructive for BPE (it strips spaces before punc
Set clean_up_tokenization_spaces=False to suppress this warning, or se
p_tokenization_spaces_for_bpe_even_though_it_will_corrupt_output=True t
leanup anyway.
Solution:
Technical engineering question: what is artificial intelligence
Solution: artificial intelligence is the ability to predict and adapt t
events, and to predict future events using the latest technology. In t
the AI is not a computer program, just a program that knows how to lea
skill or use a new technology.
If you are a tech enthusiast or a scientist working in technical engine
d suggest you read this question in a new paper that you can read today
In the computer science classroom, there are a lot of different kinds o
ns, but when they are asked, I think the most common ones are "How do y
tificial intelligence to solve a problem?", for example "Can you use AI
a problem on a computer?"
I think the most common answer to these questions is "yes." If you are
d in this, here's the paper, which is a short-term introduction to AI:
Computer science and AI
I'm not an expert on AI, but I have a broad idea that machine learning,
artificial intelligence that comes with this approach, can be used to s
lems.
This paper is part of the AI training program to help you improve your
ills, so if you would like to learn more about AI, this is a great trai
rtunity.
For
```

3 Python code:

```
from diffusers import StableDiffusionPipeline
import torch
model = StableDiffusionPipeline.from_pretrained(
    "runwayml/stable-diffusion-v1-5"
)
model = model.to("cpu")
prompt = "A modern suspension bridge over a river, engineering design"
image = model(prompt).images[0]
image.save("engineering_bridge.png")
print("Image generated successfully.")
```

Output:



4 Python code:

```
from diffusers import StableDiffusionPipeline

import torch

model = StableDiffusionPipeline.from_pretrained(

    "runwayml/stable-diffusion-v1-5"

)

model = model.to("cpu")

prompts = [

    "A modern suspension bridge",

    "A futuristic robotic arm",

    "A large engineering construction site"

]

for i, prompt in enumerate(prompts):

    image = model(prompt).images[0]

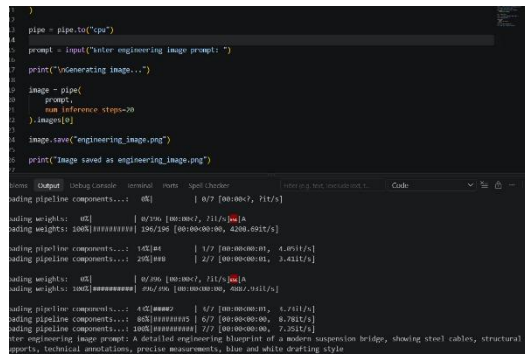
    filename = "engineering_image_" + str(i + 1) + ".png"

    image.save(filename)

    print("Generated:", filename)

print("All images generated.")
```

Output:



```
1 )
2 pipe = pipe.to("cpu")
3
4 prompt = input("Enter engineering image prompt: ")
5
6 print("\nGenerating image...\n")
7
8 image = pipe(
9     prompt,
10    run_inference_steps=50
11 ).images[0]
12
13 image.save("engineering_image.png")
14
15 print("Image saved as engineering_image.png")
16
17
18 home | Output | Debug Console | Terminal | Run | Spell Checker | Generating Next Incremental... | Code
19
20 loading pipeline components...: 0% | 0/7 [00:00:00, 7117/s]
21
22 loading weights: 0% | 0/796 [00:00:00, 2117/s]
23 loading pipeline components...: 100% [00:00:00:00, 4208.6917/s]
24
25 loading pipeline components...: 0% | 0/7 [00:00:00:00, 4.4617/s]
26 loading pipeline components...: 100% [00:00:00:00, 1.4217/s]
27
28 loading weights: 0% | 0/796 [00:00:00, 2117/s]
29 loading pipeline components...: 100% [00:00:00:00, 4002.7117/s]
30
31 loading pipeline components...: 0% [00:00:00:00, 4.4617/s]
32 loading pipeline components...: 100% [00:00:00:00, 0.7817/s]
33 loading pipeline components...: 100% [00:00:00:00, 7.7517/s]
34
35 the engineering image prompt: A detailed engineering blueprint of a modern suspension bridge, showing steel cables, structural
36 supports, technical annotations, precise measurements, blue and white drafting style
```

5 Python code:

```
import speech_recognition as sr

recognizer = sr.Recognizer()

with sr.Microphone() as source:

    print("Speak your engineering question:")

    audio = recognizer.listen(source)

try:

    text = recognizer.recognize_google(audio)

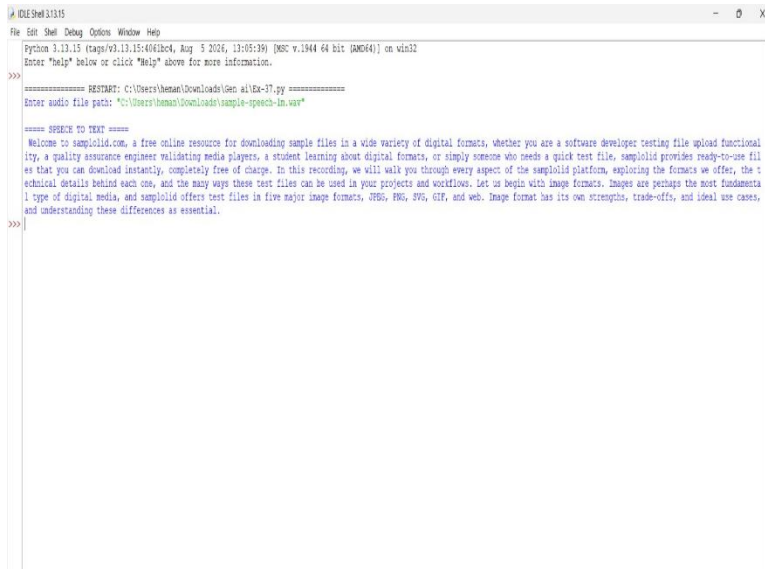
    print("Recognized Text:")

    print(text)

except:

    print("Sorry, speech could not be recognized.")
```

Output:



```
Python 3.13.15 (tags/v3.13.15:45f1bc4, Aug 5 2026, 13:05:38) [MSC v.1944 64 bit (AMD64)] on win32
Error "help" below or click "Help" above for more information.

>>>
===== RESTART: C:\Users\heman\Downloads\gen ai\Ex-37.py =====
Enter audio file path: "C:\Users\heman\Downloads\sample-speech-in.wav"

===== SPEECH TO TEXT =====
Welcome to sampleid.com, a free online resource for downloading sample files in a wide variety of digital formats, whether you are a software developer testing file upload functionality, a quality assurance engineer validating media players, a student learning about digital formats, or simply someone who needs a quick test file. sampleid provides ready-to-use files that you can download instantly, completely free of charge. In this recording, we will walk you through every aspect of the sampleid platform, exploring the formats we offer, the technical details behind each one, and the many ways these test files can be used in your projects and workflows. Let us begin with image formats. Images are perhaps the most fundamental type of digital media, and sampleid offers test files in five major image formats, JPEG, PNG, SVG, GIF, and web. Image format has its own strengths, trade-offs, and ideal use cases, and understanding these differences is essential.

>>>
```

6 Python code:

```
import pyttsx3

engine = pyttsx3.init()

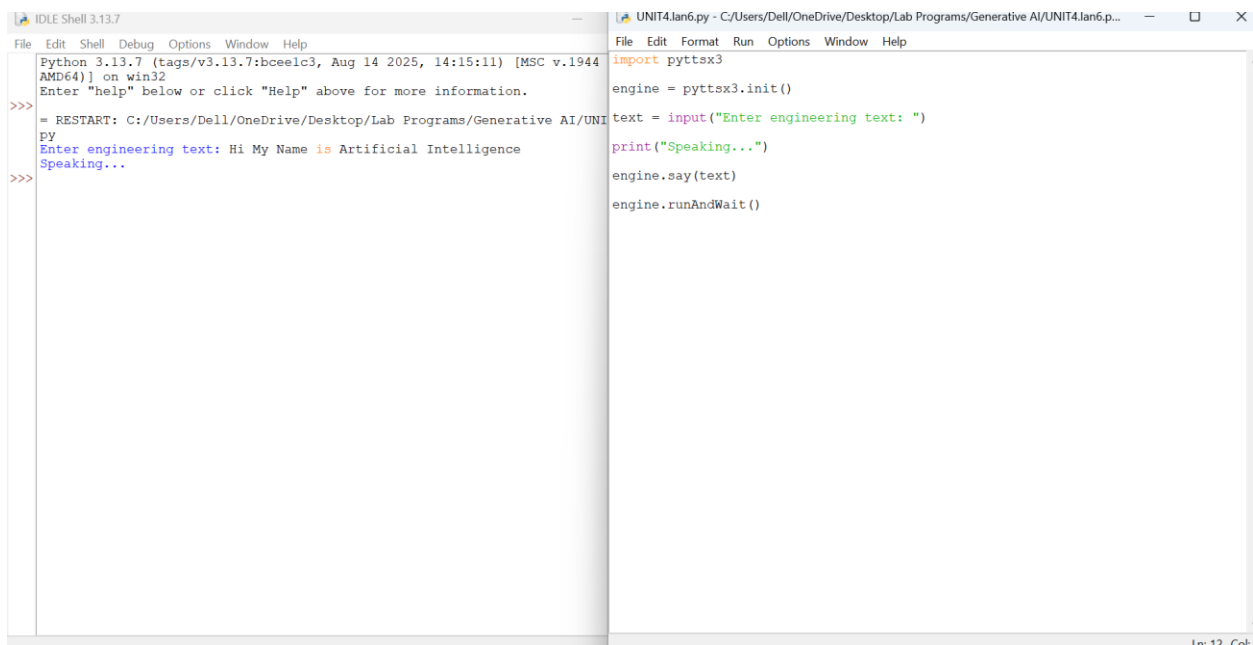
text = input("Enter engineering text: ")

print("Speaking...")

engine.say(text)

engine.runAndWait()
```

Output:



The screenshot displays two windows from a Python IDE. The left window, titled 'IDLE Shell 3.13.7', shows the execution of the code. It starts with a prompt 'Enter "help" below or click "Help" above for more information.' followed by a 'RESTART' message. The user enters 'Hi My Name is Artificial Intelligence' in response to the 'Enter engineering text:' prompt. The program then outputs 'Speaking...'. The right window, titled 'UNIT4.lan6.py - C:/Users/Dell/OneDrive/Desktop/Lab Programs/Generative AI/UNIT4.lan6.p...', shows the source code of the program, which is identical to the code block provided above. The code imports the 'pyttsx3' module, initializes an engine, takes user input, prints a message, and uses the engine to say the text and wait for it to finish.

```
Python 3.13.7 (tags/v3.13.7:bceelc3, Aug 14 2025, 14:15:11) [MSC v.1944
AMD64] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/Dell/OneDrive/Desktop/Lab Programs/Generative AI/UNI
PY
Enter engineering text: Hi My Name is Artificial Intelligence
Speaking...
>>>
```

```
UNIT4.lan6.py - C:/Users/Dell/OneDrive/Desktop/Lab Programs/Generative AI/UNIT4.lan6.p...
File Edit Format Run Options Window Help
import pyttsx3
engine = pyttsx3.init()
text = input("Enter engineering text: ")
print("Speaking...")
engine.say(text)
engine.runAndWait()
```

7 Python code:

```
from transformers import pipeline

summarizer = pipeline("summarization")

text = """

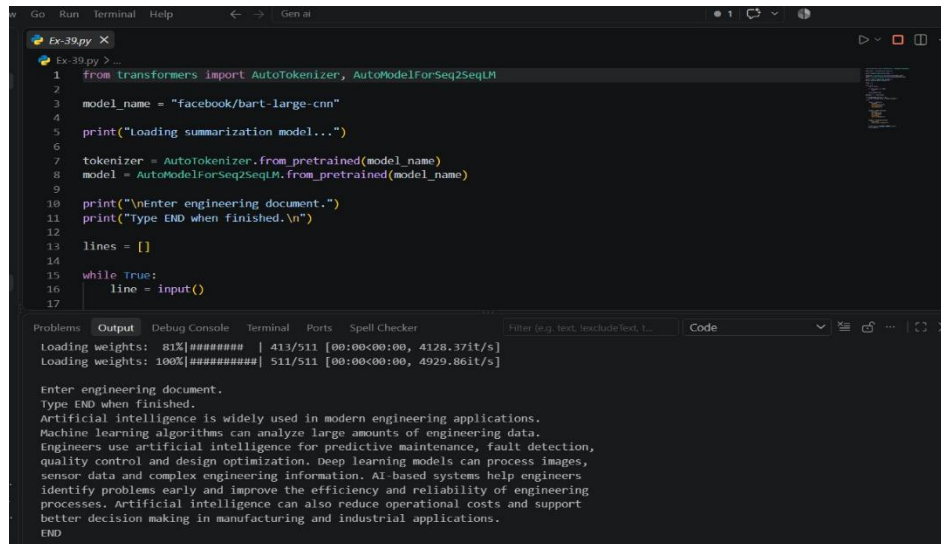
Artificial Intelligence is an important technology used in engineering.
It helps engineers analyze large amounts of data and solve complex problems.
Machine learning can be used for prediction, automation and optimization.
AI is also used in robotics, manufacturing, transportation and construction.
These technologies improve efficiency, reduce errors and save time.

"""

summary = summarizer(
    text,
    max_length=50,
    min_length=20,
    do_sample=False
)

print("Original Text:")
print(text)
print("\nSummary:")
print(summary[0]["summary_text"])
```

Output:



The screenshot shows a PyCharm IDE window with a Python script named `Ex-39.py` and its output in the `Output` pane.

Code in `Ex-39.py`:

```
1 from transformers import AutoTokenizer, AutoModelForSeq2SeqLM
2
3 model_name = "facebook/bart-large-cnn"
4
5 print("Loading summarization model...")
6
7 tokenizer = AutoTokenizer.from_pretrained(model_name)
8 model = AutoModelForSeq2SeqLM.from_pretrained(model_name)
9
10 print("\nEnter engineering document.")
11 print("Type END when finished.\n")
12
13 lines = []
14
15 while True:
16     line = input()
17
```

Output:

```
Loading weights: 81%|##### | 413/511 [00:00<00:00, 4128.37it/s]
Loading weights: 100%|#####| 511/511 [00:00<00:00, 4929.86it/s]

Enter engineering document.
Type END when finished.
Artificial intelligence is widely used in modern engineering applications.
Machine learning algorithms can analyze large amounts of engineering data.
Engineers use artificial intelligence for predictive maintenance, fault detection,
quality control and design optimization. Deep learning models can process images,
sensor data and complex engineering information. AI-based systems help engineers
identify problems early and improve the efficiency and reliability of engineering
processes. Artificial intelligence can also reduce operational costs and support
better decision making in manufacturing and industrial applications.
END
```

8 Python code:

```
from transformers import pipeline

translator = pipeline(

    "translation",

    model="Helsinki-NLP/opus-mt-en-hi"

)

text = "Artificial Intelligence is used in engineering."

result = translator(text)

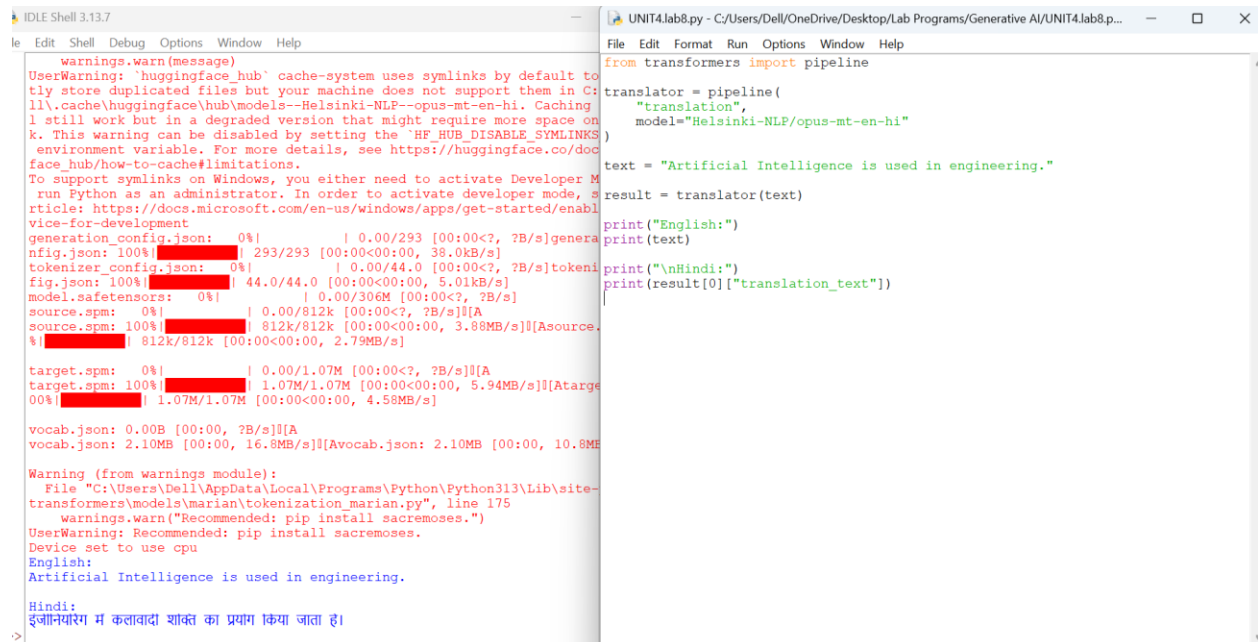
print("English:")

print(text)

print("\nHindi:")

print(result[0]["translation_text"])
```

Output:



```
IDLE Shell 3.13.7
File Edit Shell Debug Options Window Help
warnings.warn(message)
UserWarning: 'huggingface_hub' cache-system uses symlinks by default to
tly store duplicated files but your machine does not support them in C:
ll\cache\huggingface\hub\models--Helsinki-NLP--opus-mt-en-hi. Caching
l still work but in a degraded version that might require more space on
k. This warning can be disabled by setting the `HF_HUB_DISABLE_SYMLINKS`
environment variable. For more details, see https://huggingface.co/doc
face_hub/how-to-cache#limitations.
To support symlinks on Windows, you either need to activate Developer M
run Python as an administrator. In order to activate developer mode, s
rticle: https://docs.microsoft.com/en-us/windows/apps/get-started/enab
vice-for-development
generation_config.json: 0%|          | 0.00/293 [00:00<?, ?B/s]genera
nfig.json: 100%|██████████| 293/293 [00:00<00:00, 38.0kB/s]
tokenizer_config.json: 0%|          | 0.00/44.0 [00:00<?, ?B/s]tokeni
fig.json: 100%|██████████| 44.0/44.0 [00:00<00:00, 5.01kB/s]
model.safetensors: 0%|          | 0.00/306M [00:00<?, ?B/s]
source.spm: 0%|          | 0.00/812k [00:00<?, ?B/s][A
source.spm: 100%|██████████| 812k/812k [00:00<00:00, 3.88MB/s][Asource.
%|██████████| 812k/812k [00:00<00:00, 2.79MB/s]

target.spm: 0%|          | 0.00/1.07M [00:00<?, ?B/s][A
target.spm: 100%|██████████| 1.07M/1.07M [00:00<00:00, 5.94MB/s][Atarge
00%|██████████| 1.07M/1.07M [00:00<00:00, 4.58MB/s]

vocab.json: 0.00B [00:00, ?B/s][A
vocab.json: 2.10MB [00:00, 16.8MB/s][Avocab.json: 2.10MB [00:00, 10.8MB/s]

Warning (from warnings module):
File "C:\Users\Dell\AppData\Local\Programs\Python\Python313\Lib\site-
transformers\models\marian\tokenization_marian.py", line 175
warnings.warn("Recommended: pip insTall sacremoses.")
UserWarning: Recommended: pip install sacremoses.
Device set to use cpu
English:
Artificial Intelligence is used in engineering.

Hindi:
इंजीनियरिंग में कलावादी शक्ति का प्रयोग किया जाता है।
>
```

9 Python code:

```
import re

job_description = """
We need an engineer with Python, SQL, Machine Learning
and NLP skills with at least 2 years of experience.
"""

resumes = {
    "Rahul": """
Python developer with 3 years of experience.
Skills: Python, SQL, Machine Learning, NLP.
""",
    "Deepa": """
Software developer with 1 year of experience.
Skills: Java, HTML and CSS.
""",
    "Arun": """
Data analyst with 4 years of experience.
Skills: Python, SQL and Machine Learning.
"""
}

for name, resume in resumes.items():
    score = 0
    if "python" in resume.lower():
        score += 1
    if "sql" in resume.lower():
        score += 1
    if "machine learning" in resume.lower():
        score += 1
```

```

if "nlp" in resume.lower():
    score += 1

years = re.findall(r'(\d+)\s*years?', resume.lower())
experience = int(years[0]) if years else 0
if experience >= 2:
    score += 2

print("\nCandidate:", name)
print("Experience:", experience, "years")
print("Score:", score)
if score >= 4:
    print("Status: Selected")
else:
    print("Status: Not Selected")

```

Output:

The image shows two side-by-side windows from a code editor. The left window, titled 'IDLE Shell 3.13.7', displays the output of a Python script. It shows three candidates: Rahul (3 years experience, score 6, status: Selected), Deepa (1 year experience, score 0, status: Not Selected), and Arun (4 years experience, score 5, status: Selected). The right window, titled 'UNIT4.lab9.py - C:/Users/Dell/OneDrive/Desktop/Lab Programs/Generative AI/UNIT4.lab9.p...', shows the source code of the script. The code defines a job description, a list of resumes, and a function to calculate a score based on keywords ('python', 'sql', 'machine learning', 'nlp') and years of experience. It then iterates over the resumes and prints the results for each candidate.

```

IDLE Shell 3.13.7
Python 3.13.7 (tags/v3.13.7:bceelc3, Aug 14 2025, 14:15:11) [MSC v.1916 AMD64] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/Dell/OneDrive/Desktop/Lab Programs/Generative AI/Py
Candidate: Rahul
Experience: 3 years
Score: 6
Status: Selected

Candidate: Deepa
Experience: 1 years
Score: 0
Status: Not Selected

Candidate: Arun
Experience: 4 years
Score: 5
Status: Selected
>>>

UNIT4.lab9.py - C:/Users/Dell/OneDrive/Desktop/Lab Programs/Generative AI/UNIT4.lab9.p...
import re

job_description = """
We need an engineer with Python, SQL, Machine Learning
and NLP skills with at least 2 years of experience.
"""

resumes = {
    "Rahul": """
Python developer with 3 years of experience.
Skills: Python, SQL, Machine Learning, NLP.
""",
    "Deepa": """
Software developer with 1 year of experience.
Skills: Java, HTML and CSS.
""",
    "Arun": """
Data analyst with 4 years of experience.
Skills: Python, SQL and Machine Learning.
"""
}

for name, resume in resumes.items():
    score = 0

    if "python" in resume.lower():
        score += 1

    if "sql" in resume.lower():
        score += 1

    if "machine learning" in resume.lower():
        score += 1

    if "nlp" in resume.lower():
        score += 1

```

10 Python code:

```
from transformers import pipeline

generator = pipeline("text-generation", model="gpt2")

topic = input("Enter research topic: ")

prompt = """

Research Topic: """ + topic + """

Provide:

1. Introduction

2. Important keywords

3. Short summary

4. Applications

"""

result = generator(

    prompt,

    max_length=150,

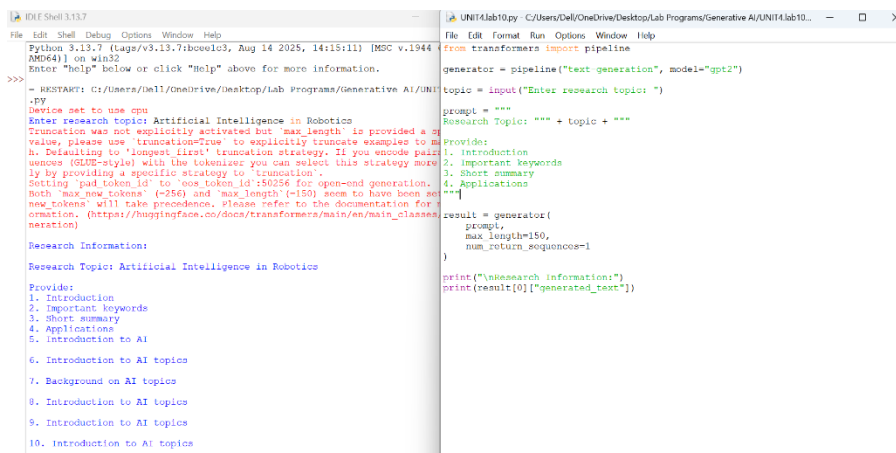
    num_return_sequences=1

)

print("\nResearch Information:")

print(result[0]["generated_text"])

Output:
```



```
Python 3.13.7 (tags/v3.13.7:bee51c3, Aug 14 2025, 14:15:11) [MSC v.1944
AMD64] on win32
Enter "help" below or click "Help" above for more information.
>>>
- RESTART: C:/Users/Dell/OneDrive/Desktop/Lab Programs/Generative AI/UNIT4/lab10.py
Device set to use cpu
Enter research topic: Artificial Intelligence in Robotics
Truncation was not explicitly activated but 'max_length' is provided a value, please use 'truncation=True' to explicitly truncate examples to max length. Defaulting to 'longest first' truncation strategy. If you encode pairs of sequences (GLUE-style) with the tokenizer you can select this strategy more precisely by providing a specific strategy to 'truncation'.
Setting 'pad_token_id' to 'eos_token_id':50256 for open-end generation.
Both 'max_new_tokens' (=256) and 'max_length' (=150) seem to have been set, 'max_new_tokens' will take precedence, please refer to the documentation for more information. (https://huggingface.co/docs/transformers/main/en/main_classes_generation)

Research Information:

Research Topic: Artificial Intelligence in Robotics

Provide:
1. Introduction
2. Important keywords
3. Short summary
4. Applications
5. Introduction to AI
6. Introduction to AI topics
7. Background on AI topics
8. Introduction to AI topics
9. Introduction to AI topics
10. Introduction to AI topics
```